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Wireless charging of a wrist-mounted sensor platform

Abstract

Wearable devices are described herein including a housing and a mount configured to mount a contact surface of the housing to an external surface of a wearer. The wearable devices further include a coil disposed in the housing proximate to the contact surface and at least one sensor disposed on the contact surface and configured to detect one or more properties of the body of the wearer. The wearable devices are powered by a rechargeable battery disposed within the wearable devices. The wearable devices additionally include a recharger disposed within the wearable devices and configured to recharge the rechargeable battery using electromagnetic energy received by the coil. The sensor is disposed within a central portion of the contact surface enclosed by the coil.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/665,931, filed Oct. 28, 2019, which is a continuation of U.S. patent application Ser. No. 14/303,865, filed Jun. 13, 2014. The foregoing applications are incorporated herein by reference.

BACKGROUND

- (1) Unless otherwise indicated herein, the materials described in this section are not prior art to the claims in this application and are not admitted to be prior art by inclusion in this section.
- (2) A number of scientific methods have been developed to detect and/or measure one or more physiological properties of a human. Some of these methods can be implemented in the form of portable, low-power sensing devices. Such sensing devices can be included in wearable devices that can be mounted to a human body to enable continuous sensing detection and/or measurement of the one or more physiological properties.
- (3) In some examples, the one or more physiological properties include properties of analytes in the human body. The one or more analytes could be any analytes that, when present in or absent from a person's body, or present at a particular concentration or range of concentrations, may be indicative of a medical condition or health state of the person. The one or more analytes could be substances whose distribution, action, or other properties, interactions, or activities throughout a human's body was of scientific, medical, personal, or other interest.
- (4) In some examples, the one or more physiological properties include a Galvanic skin response of skin of the human body. The Galvanic skin response is a change in the conductivity and/or electrical potential of the skin due to changes in the moisture level of the skin. This change in

moisture level can be caused by activation or inactivation of sweat glands in the skin. The Galvanic skin response includes the Galvanic skin resistance (GSR), a measure of the conductivity of the skin between two or more points, and the Galvanic skin potential (GSP), a measure of the voltage difference between two or more points on the skin

SUMMARY

(5) Some embodiments of the present disclosure provide a wearable device including: (i) a housing; (ii) a rechargeable battery disposed within the wearable device; (iii) a mount configured to mount the housing to an external body surface of a body such that a contact surface of the housing is in contact with the external body surface; (iv) a coil configured to receive electromagnetic energy, wherein the coil is disposed within the housing and proximate to the contact surface of the housing, wherein the coil comprises a plurality of windings surrounding an interior, and wherein the contact surface of the housing includes a central portion proximate to the interior of the coil and outlined by the windings of the coil; (v) a sensor disposed on the contact surface of the housing in the central portion, wherein the sensor is configured to detect one or more properties of the body via the external body surface; and (vi) electronics disposed in the wearable device, wherein the electronics comprises: (a) a recharger configured to recharge the rechargeable battery, wherein the recharger is configured to be powered by electromagnetic energy received by the coil; and (b) a controller configured to operate the sensor to detect one or more properties of the body via the external body surface.

(6) These as well as other aspects, advantages, and alternatives, will become apparent to those of ordinary skill in the art by reading the following detailed description, with reference where appropriate to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A is a schematic illustration of elements of an example wearable device.
- (2) FIG. 1B is a cross-sectional schematic of the example wearable device illustrated in FIG. 1A mounted to an example wireless charger.
- (3) FIG. 1C is side partial cross-sectional view of the example wearable device illustrated in FIG. 1A, while on a human wrist.
- (4) FIG. 2A is a cross-sectional schematic of an example wearable device.
- (5) FIG. 2B is an illustration of elements of the example wearable device illustrated in FIG. 2A.
- (6) FIG. 2C is an illustration of a reverse view of the elements illustrated in FIG. 2B.
- (7) FIG. 2D is an illustration of a surface of the example wearable device illustrated in FIG. 2A.
- (8) FIG. 3 is a perspective view of an example wearable device.
- (9) FIG. 4A is a perspective top view of an example wrist-mounted device, when mounted on a wearer's wrist.
- (10) FIG. 4B is a perspective bottom view of an example wrist-mounted device shown in FIG. 4A, when mounted on a wearer's wrist.
- (11) FIG. 5 is a block diagram of an example system that includes a plurality of wearable devices in communication with a server.
- (12) FIG. 6 is a functional block diagram of an example device.

DETAILED DESCRIPTION

(13) In the following detailed description, reference is made to the accompanying figures, which form a part hereof. In the figures, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative embodiments described in the detailed description, figures, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the scope of the subject matter presented herein. It

will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are explicitly contemplated herein.

(14) Further, while embodiments disclosed herein make reference to use on or in conjunction with a living human body, it is contemplated that the disclosed methods, systems and devices may be used in any environment where the operation of a sensor is desired, where the sensor is powered by a rechargeable battery that can be recharged using electromagnetic energy received by a coil disposed proximate to the sensor. The environment may be any living or non-living body or a portion thereof, a gel, an emulsion, a fluid conduit, a fluid reservoir, etc. For example, one of skill in the art will recognize that the embodiments disclosed herein may be used to sense analytes present in a water treatment system.

I. Overview

(15) A wearable device may be configured to measure one or more physiological parameters of a wearer. The one or more physiological parameters can include galvanic skin response, pulse timing and/or rate, blood oxygenation, temperature, the concentration of one or more analytes in the blood, or some other parameters. To measure the one or more physiological parameters, the wearable device may include a sensor disposed in a housing of the device so as to have access to (e.g., to contact, to be able to receive light from, to be able to illuminate) the wearer's skin at a location such as the wearer's wrist, forearm, upper arm, leg, thigh, etc. With the sensor having access to the wearer's skin, electronics within the device may be used to operate the sensor to measure the one or more physiological parameters. The electronics may be powered by a rechargeable battery in the wearable device. The wearable device may further include a recharger for recharging the rechargeable battery. To recharge the rechargeable battery, the recharger may be connected to a coil configured to receive electromagnetic energy from an external source (e.g., a wireless charger).

(16) In one example, the wearable device includes a housing (e.g., a water-proof housing) and a mount (e.g., a band) that can mount the housing on a particular body location, such as a wrist. To maximize the amount of electromagnetic energy the coil can receive, the coil can be disposed along an outer edge of the housing. The sensor can be disposed in the housing and within the extent of the coil in order to access the wearer's skin. Electronics disposed in the housing may include a reference voltage sources, current sources, amplifiers, comparators, ADCs, or other elements configured to operate the sensor to detect the one or more physiological parameters of the wearer. The electronics may also include the recharger. The recharger could be configured to operate the coil to receive electromagnetic energy of a specific frequency (e.g., to have a capacitance related to an inductance or other properties of the coil; to adjust an effective capacitance of the recharger to change the specific frequency and/or to adapt to one or more properties of the environment of the wearable device). The recharger could additionally be configured to operate the coil or other components to communicate with a wireless charger.

(17) The wearable device could include one or more magnetic shield elements to shield components of the wearable device from electromagnetic energy (e.g., energy from a wireless charger). That is, certain components of the wearable device could experience heating or other effects when exposed to a charging electromagnetic energy (e.g., a radio-frequency EM field). Further, the one or more magnetic shield elements could be configured to increase the efficiency of energy transfer to the coil. In some examples, the one or more magnetic shield elements include a sheet composed of ferrite material or of some other material having a specified electromagnetic property (e.g., a specified low magnetic reluctance, a specified high resistivity) disposed between the coil and other elements of the wearable device (e.g., the rechargeable battery, the recharger, other electronics). In some examples, the coil and sensor are disposed on a first printed circuit (e.g., a flexible printed circuit) and connected to elements on the other side of the magnetic shield (e.g., the rechargeable battery, the recharger, other electronics) by a flexible interconnect that passes through a slot, hole, or other feature of the magnetic shield. In some examples, the sensor is

disposed on a side of the magnetic shield opposite the coil and accesses (e.g., illuminates, detects light or other energy received from) the external body surface of a wearer through a hole or other feature of the magnetic shield. In some examples, the printed circuit on the coil side of the magnetic shield includes a minimal amount of metal (e.g., conductive circuit traces and/or interconnects) to reduce heating of the wearable device and/or to increase the efficiency of energy transfer to the coil. For example, the printed circuit on the coil side of the magnetic shield could lack a ground plane.

(18) In some examples, the wearable device may include a user interface that is configured to provide user-discernible indications (e.g., visual, audible, and/or tactile indications) of one or more physiological parameters measured by the device, such as galvanic skin response, temperature, or pulse rate. In some examples, the wearable device may include a wireless communication interface that can transmit data to an external device, for example, using Bluetooth, ZigBee, WiFi, or other wireless communication protocol. The data transmitted by the wireless communication interface may include data indicative of one or more physiological parameters measured by the device.

II. Example Wearable Sensor Platforms

(19) Wearable devices as described herein can be configured to be mounted to an external body surface of a wearer and to enable a variety of applications and functions. Such wearable devices include a housing (i.e., a rigid or semi-rigid enclosure) and a mount configured to mount a contact surface of the housing to the external body surface of the wearer. A sensor disposed on a central portion of the contact surface can detect one or more properties of the body of the wearer when the contact surface is mounted to the external body surface. A coil configured to receive electromagnetic energy is disposed within the housing, proximate to the contact surface, and windings of the coil outline the central portion of the contact surface. Such wearable devices could enable a variety of applications, including measuring physiological information about a wearer, indicating such measured physiological information or other information to the wearer (e.g., using a vibrator, a screen, a beeper), or other functions. Energy received using the coil could be used to power the wearable device and/or to recharge a rechargeable battery of the wearable device that is configured to power the wearable device.

(20) FIG. 1A illustrates a schematic view of components of an example wearable device **100**. The wearable device includes a housing **110** (e.g., a water-resistant and/or water-proof housing) configured to contain electronic components and to be mounted to an external body surface of a wearer by a mount **120** (e.g., a band). The mount **120** is a band configured to enclose a wrist of a human and to mount a contact surface **130** of the housing **110** in contact with the wrist of the wearer. A sensor **140** is disposed on a central portion of the contact surface and a coil **150** is disposed within the housing **110** proximate to the contact surface **130**. The coil **150** comprises windings that outline the central portion of the contact surface **130** such that the central portion of the contact surface **130** is proximate to the interior of the coil **150**. The wearable device **100** includes additional elements that are not shown, e.g., electronics configured to operate the coil **150** and/or sensor **140** and to enable applications and/or functions of the wearable device **100**, a rechargeable battery configured to power the wearable device **100**, a recharger configured to recharge the rechargeable battery using electromagnetic energy received using the coil **150**, or other components. Components of the wearable device **100** could be disposed on or within the housing **110**, the mount **120**, or some other elements of the wearable device **100** (not shown); e.g., a second housing.

(21) The housing **110** could be configured to be water-resistant and/or water-proof. That is, the housing **110** could be configured to include sealants, adhesives, gaskets, welds, press-fitted seams, and/or other joints such that the housing **110** was resistant to water entering an internal volume or volumes of the housing **110** when the housing **110** is exposed to water. The housing **110** could further be water-proof, i.e., resistant to water entering an internal volume or volumes of the housing **110** when the housing **110** is submerged in water. For example, the housing **110** could be water-

proof to a depth of 1 meter, i.e., configured to resist water entering an internal volume or volumes of the housing **110** when the housing **110** is submerged to a depth of 1 meter. Further, the interface between the housing **110** and other elements of the wearable device **100** (e.g., elements of a sensor, buttons, user interface elements, electrical contacts) protruding from, embedded in the surface of, or otherwise interrupting the material of the housing **110** could be configured such that the combination of the housing **110** and the other elements of the wearable device **100** is water-resistant and/or water-proof.

(22) FIG. **1B** is a cross-sectional view of the wearable device **100** mounted on a wireless charger **160**. The wireless charger **160** includes a charging coil **170** configured to emit electromagnetic energy. The wearable device **100** can be mounted on (e.g., placed on, secured to, disposed in proximity to, aligned with) the wireless charger **160** such that the contact surface **130** of the wearable device **100** is in contact with a charging surface **180** of the wireless charger **160**.

(23) In some embodiments, the wearable device **100** and/or wireless charger **160** could be configured to facilitate efficient transfer of electromagnetic energy between the charging coil **170** of the wireless charger **160** and the coil **150** of the wearable device **100** by aligning, ensuring proximity of, or effecting some other specified relative arrangement between the coil **150** and the charging coil **170**. For example, the wearable device **100** and/or wireless charger **160** could include elements and/or be configured to ensure alignment between the coil **150** and charging coil **170**, e.g., by the contact surface **130** and the charging surface **180** having matching and/or interlocking shapes, by including one or more permanent magnets configured to exert aligning magnetic forces between the wearable device **100** and the wireless charger **160**, by including alignment markings to indicate to a user a proper alignment of the wearable device **100** on the wireless charger **160**, or by the addition of some other components.

(24) Further, the wearable device **100** and/or wireless charger **160** could include one or more magnetic shims or other materials having one or more specified magnetic properties to modify the transfer of electromagnetic energy between the coil **150** and the charging coil **170**. For example, the wearable device **100** could include a magnetic shield disposed proximate to the coil **150** on a side of the coil **150** opposite the contact surface **130**. The magnetic shield could be configured to ‘focus’ electromagnetic energy directed toward the coil **150** such that the coil **150** can receive more of the electromagnetic energy. The magnetic shield could additionally or alternatively be configured to shield components of the wearable device **100** from electromagnetic energy (e.g., to prevent electromagnetic energy directed toward the wearable device **100** from heating or otherwise affecting components (e.g., electronics, rechargeable batteries) opposite the magnetic shield from the direction of the electromagnetic energy).

(25) The coil **150** can be configured in a number of ways to enable efficient reception of electromagnetic energy using the coil **150** or to enable and/or facilitate a number of other applications. The windings of the coil **150** could be disposed proximate to a peripheral portion of the contact surface **130** of the housing **110** such that an area enclosed by the coil **150** (e.g., the central portion of the contact surface **130** of the housing **110**) is maximized and/or such that a separation distance between the coil **150** and the charging coil **170** is minimized. The coil could have a rectangular shape, an elliptical shape, or some other shape according to an application; for example, the shape of the coil **150** could correspond to the shape of the contact surface **130**. The coil **150**, recharger (not shown), or other components could be configured to enable efficient reception of electromagnetic energy of a specific frequency (e.g., 100 kilohertz to 200 kilohertz) by the coil. For example, the coil and a capacitor of the recharger could be configured to have a resonant frequency equal to the specific frequency of the electromagnetic energy.

(26) The wireless charger **160** could be configured in a variety of ways and include a variety of additional components to facilitate the emission of electromagnetic energy such that the coil **150** of the wearable device **100** can receive the transmitted electromagnetic energy. The wireless charger could include switches, coils, capacitors, variable frequency drives, or other electronics configured

to emit electromagnetic energy that could be received by the coil **150** of the wearable device **100**. In some examples, the wireless charger **160** could be configured to detect the presence, energy capacity, or other properties of the wearable device **100** and to emit electromagnetic energy having one or more properties related to the detected presence, energy capacity, or other property. In some examples, the wireless charger **160** could receive information from the wearable device **100** indicating an amount of electromagnetic energy to emit toward the coil **150** of the wearable device **100**. For example, the wearable device **100** could operate the coil **150** to change the impedance or some other electromagnetically detectable property of the coil **150** in a pattern related to an amount of energy that the wireless charger **160** could emit toward the coil **150** of the wearable device **100** using the charging coil **170**. In some examples, the wireless charger **160** and/or wearable device **100** could comply with one or more wireless charging standards (e.g., the Qi wireless charging standard).

(27) FIG. **1C** is a partial cross-sectional side view of a human wrist illustrating the operation of the wearable device **100** when mounted to a human wrist. In the example shown in FIG. **1C**, the wearable device **100** is mounted (using the mount **120**) such that the contact surface **130** is in contact with the posterior side **190** of the wearer's wrist (i.e., an external body surface of the wearer). Sensor **140** is positioned over a portion of the wrist where subsurface vasculature **195** is easily observable. The sensor **140** could be operated to detect one or more properties of the skin, subsurface vasculature **195**, or other elements of the wearer's body proximate to the wearable device **100** via the posterior side **190** of the wearer's wrist, e.g., by contacting, directing energy (e.g., electrical, magnetic, illumination, acoustic waves) through/into, detecting energy (e.g., electrical, magnetic, illumination, acoustic waves) received from/through, or otherwise interacting with and/or through an external body surface of the wearer (i.e., the posterior side **190** of the wearer's wrist).

(28) The sensor **140** is disposed on the central portion of the contact surface **130** of the housing **110** such that the sensor **140** can access the body of the wearer (i.e., make electrical, thermal or other physical contact with, receive light or other electromagnetic energy from, emit light or other electromagnetic energy toward, detect and/or emit an electric, magnetic, or other field from/to the body of the wearer) without being occluded or otherwise blocked by the coil **150**. The sensor **140** accessing the body of the wearer could include the sensor being disposed outside the housing, inside the housing, proximate to a window or other hole in the housing, proximate to a filter or other element permitting partial access to the body of the wearer, or some other configuration.

(29) The sensor **140** could include a light sensor, an IR sensor, an electric field sensor, a magnetic field sensor, an electromagnetic energy sensor, a temperature sensor, an electric current sensor, an electric potential sensor, or some other sensor or combination of sensors. The sensor **140** could include a light emitter, an IR emitter, and electromagnetic energy emitter, a heater, a vibrator, or some other energy emitter. The sensor **140** could be configured to operate in direct contact with an external body surface of the wearer (e.g., configured to include a heat- or electricity-conducting probe or other element in physical contact with the skin of the wearer to facilitate detection of one or more properties of the body of the wearer). The sensor **140** could be configured to detect a Galvanic skin resistance of skin at the external body surface of the wearer by applying an electric potential between two probes in electrical contact with skin at the external body surface of the wearer.

(30) The sensor **140** could interact with and/or detect one or more properties of specific elements or components of the body of the wearer. In some examples, the sensor **140** could act to illuminate or otherwise direct energy toward elements in the body of the wearer and could detect a light or other energy emitted by, reflected by, scattered by, or otherwise received from the elements in response to the illumination and/or direction of energy toward the elements. For example, the sensor **140** could be configured to illuminate a fluorophore, chromophore, or other optic chemical, moiety, analyte, or other element of the body of the wearer and to detect light emitted, scattered, reflected,

or otherwise received from the element of the body of the wearer in response to the illumination. One or more properties of the illumination and/or of the detected light could be used to determine one or more properties of the body of the wearer. For example, a degree of scattering of light having a specified wavelength could be used to determine a pulse rate of the wearer, a blood oxygenation of blood in the skin and/or subsurface vasculature **195** of the wearer, or some other property of the body of the wearer.

(31) In some examples, the coil **150** could be operated to facilitate some application of the sensor **140**. For example, the coil **150** could be operated to emit an electromagnetic energy having a specified frequency when the wearable device **100** is mounted to the external body surface of the wearer. The electromagnetic energy could excite, apply a force to, or otherwise interact with some element of the body of the wearer such that the sensor **140** could detect one or more properties of the body of the wearer. Other methods of operation of the coil **150** to enable detection of one or more properties of the body of the wearer are anticipated.

(32) In some examples, the body of the wearer (e.g., the subsurface vasculature **195**) could include artificial or other contrast agents (e.g., fluorophores, fluorescent nanodiamonds, chromophores, acoustic particles, magnetic particles) functionalized or otherwise configured to enable the detection of one or more properties of the body of the wearer using the sensor **140**. For example, a contrast agent including a fluorophore could be configured to selectively bind to an analyte of interest in the blood of the wearer, and sensor **140** could be operated to determine to presence, location, binding state, or other properties of the contrast agent in the blood. The determined one or more properties of the contrast agent could be used to determine the presence or concentration of the analyte in the blood of the wearer. Other contrast agents, properties of the body of the wearer, and configurations and method of operation of the sensor **140** are anticipated.

(33) Wearable devices and other embodiments as described herein can include a variety of components configured in a variety of ways. FIG. 2A illustrates a cross-sectional schematic view of a wearable device **200** that includes a housing **210** (e.g., a water-resistant and/or water-proof housing) and mount (e.g., a band **220**, as shown in FIG. 2D) configured to mount a contact surface **230** of the housing to an external body surface (e.g., a wrist) of a wearer. The housing **210** contains a coil **250** disposed within the housing **210** proximate to the contact surface **230** that is configured to receive electromagnetic energy. The wearable device **200** additionally includes a variety of sensors and components of sensors (e.g., **240**, **242**, **244**) disposed on the contact surface **230** and configured to detect one or more properties of the body of the wearer. FIGS. 2B and 2D illustrate other sensors and components of sensors (**246**, **248a**, **248b**, **249a**, **249b**) included in the wearable device **200**.

(34) The sensors and components of sensors are mounted on a flexible printed circuit board (PCB) **260** that is mounted onto a magnetic shield **270**. The coil **250** is also disposed on the magnetic shield **270** and is electronically coupled to the flexible PCB **260**. A flexible interconnect of the flexible PCB **260** passes through a slot **275** in the magnetic shield **270** and includes a first connector **265** that is connected to a second connector **284** that is disposed on a circuit board **280** on the opposite side of the magnetic shield **270** from the coil **250** and the sensors and components of sensors (**240**, **242**, **244**, **246**, **248a**, **248b**, **249a**, **249b**). Electronics **282** are also disposed on the circuit board **280**. An interface **290** and a rechargeable battery **286** are operatively coupled to the circuit board **280** and disposed within the housing **210** on the same side of the magnetic shield **270** as the circuit board **280** and electronics **282** disposed thereupon.

(35) The electronics **282** could include a variety of different components configured in a variety of ways to enable applications of the wearable device. The electronics **282** could include controllers, amplifiers, switches, display drivers, touch sensors, wireless communications chipsets (e.g., Bluetooth radios or other radio transceivers and associated baseband circuitry to enable wireless communications between the wearable device **200** and some other system(s)), or other components. The electronics **282** include a controller configured to operate one or more sensors and/or

components of sensors (e.g., **240**, **242**, **244**, **246**, **248a**, **248b**, **249a**, **249b**) to detect one or more properties of the body of the wearer. The controller could include a processor configured to execute computer-readable instructions (e.g., program instructions stored in data storage of the wearable device **200**) to enable applications of the wearable device **200**. The electronics **282** additionally includes a recharger that is configured to recharge the rechargeable battery **286** and that is configured to be powered by electromagnetic energy received by the coil **250** (i.e., the recharger is configured to recharge the rechargeable battery **286** using energy received by the coil **250**). The electronics **282** can include additional or alternative components according to an application of the wearable device **200**.

(36) The rechargeable battery **286** is configured to power the wearable device **200** using stored electrochemical energy and to be recharged a plurality of times. The rechargeable battery **286** could include one or more of a variety of rechargeable battery chemistries, including lead-acid, nickel-metal-hydride, nickel-cadmium, lithium-ion, lithium-polymer, or some other rechargeable battery chemistry. The recharger of the electronics **282** could be configured to recharge the rechargeable battery **286** by applying a constant current, a constant voltage, a trickle current, or some other electrical energy having one or more specified properties to two or more electrodes of the rechargeable battery **286**. The rechargeable battery **286** could include one or more thermistors that the controller, the recharger, or some other component of the wearable device **200** could operate to determine a temperature of the rechargeable battery **286** and to prevent damage of the rechargeable battery **286** by reducing a charging rate, a discharging rate, or some other property of use of the rechargeable battery **286** to prevent damage of the rechargeable battery **286**.

(37) The interface **290** includes a display configured to present an image to a wearer and to detect one or more finger presses of a wearer on the interface **290**. The controller or some other component(s) of the electronics **282** could operate the interface **290** to provide a user interface to a wearer or other user of the device and to enable the wearer or other user to affect the operation of the wearable device **200**, to determine some property of the wearable device **200** and/or of the wearer of the wearable device **200**, or to provide some other functionality or application to the wearer and/or user. For example, the interface **290** could be operated to indicate to the wearer one or more properties of the body of the wearer (e.g., a GSR of the skin of the wearer proximate to the wearable device **200**, a pulse rate of the wearer, some other medical condition or state of the wearer). For example, the wearer could press an indicated region of the interface **290** to indicate that the wearable device **200** should begin logging detected medical information about the wearer. Other indicated information, changes in operation of the wearable device **200**, or other functions and applications of the interface **290** are anticipated.

(38) FIGS. 2B and 2C show front and back views, respectively (i.e., a views from the direction of the contact surface **230** and opposite the contact surface **230**, respectively) of elements of the wearable device **200**. Specifically, FIGS. 2B and 2C illustrate the magnetic shield **270**, flexible PCB **260**, and some of the components disposed on the magnetic shield **270** and/or flexible PCB **260** (e.g., **265**, **250**, **240**, **242**, **244**, **246**, **248a**, **248b**). The coil **250** is disposed on the magnetic shield **270** such that when the coil **250**, magnetic shield **270**, and other components are assembled into the housing **210** of the wearable device as shown in FIG. 2A, the coil **250** is proximate to the contact surface **230** of the housing **210** such that windings of the coil **250** outline a central portion of the contact surface **230** proximate to the interior of the coil **250**.

(39) Further, the sensors and/or components of sensors (e.g., **240**, **242**, **244**, **246**, **248a**, **248b**, **249a**, **249b**) are disposed on the flexible PCB **260** such that, when assembled into the housing **210** of the wearable device as shown in FIG. 2A, the sensors and/or components of sensors are disposed on the contact surface **130** in the central portion of the contact surface **130**. The flexible PCB **270** includes a flexible interconnect that passes through the slot **275** in the magnetic shield **270**. The flexible interconnect is configured to electrically couple the electronics **282** to the coil **250** and the one or more sensors and/or components of sensors (e.g., through the connectors **265**, **284**, the

circuit board **280**, and through traces patterned on the flexible interconnect). Note that, in some embodiments, the flexible interconnect could include a cable, discrete wires, a second flexible PCB electrically connected to the flexible PCB **270**, or some other element(s) configured to pass through a hole, slot, or other feature of the magnetic shield **280** and to electrically couple the coil **250** and sensors and/or components of sensors on a first side of the magnetic shield **270** to one or more electronic components on a second side of the magnetic shield **270** opposite the first side.

(40) In some embodiments, one or more sensors could be disposed on the circuit board **280** or some other location on a side of the magnetic shield **270** opposite the coil **250** and could access (e.g., illuminate, detect light or other energy received from) the external body surface of a wearer through a hole or other feature of the magnetic shield **270**.

(41) The magnetic shield **270** is composed of one or more materials having specified magnetic properties such that, when electromagnetic energy is directed toward the wearable device **200** from the direction of the contact surface **230** such that the coil **250** can receive a portion of the directed electromagnetic energy, the electronics **282**, rechargeable battery **286**, and/or other components on a side of the magnetic shield **270** opposite the coil **250** are heated less by that directed electromagnetic energy than if the magnetic shield **270** was not present. That is, the magnetic shield **270** is configured to act as a shield to protect the electronics **282**, rechargeable battery **286**, and/or other components on a side of the magnetic shield **270** opposite the coil **250** from heating, radio-frequency noise, and/or other effects of electromagnetic energy directed at the coil **250** to provide the wearable device **200** with energy (e.g., energy that can be used to recharge the rechargeable battery **286**). The magnetic shield **270** could also be configured to increase an efficiency of energy transfer between an emitter of electromagnetic energy and the coil **250**. For example, the magnetic shield **270** could be configured to increase a coupling between the coil **250** and a charging coil that is directing electromagnetic energy toward the coil **250** (e.g., by concentrating within the magnetic shield **250** magnetic flux that passes through the windings of the coil **250** such that the more of the magnetic flux acts to transfer energy to the coil **250**). For example, the magnetic shield **270** could be wider than the coil **250** by at least 0.5 millimeters.

(42) The magnetic shield **270** could include materials having a specified high permeability such that the magnetic shield **270** could redirect magnetic flux to reduce heating of the electronics **282** and/or rechargeable battery **286** due to electromagnetic energy directed toward the coil **250** and/or to increase the efficiency of energy transfer to the coil **250** from electromagnetic energy directed toward the coil **250** (e.g., by a charging coil of a wireless charger). The magnetic shield **270** could include materials having a specified low electrical conductivity such that the magnetic shield **270** is minimally heated by exposure to time-varying electromagnetic fields (i.e., the magnetic shield **270** could experience minimal eddy currents when exposed to time-varying magnetic fields). The magnetic shield **270** could include soft magnetic materials, zinc ferrite, alpha iron, iron oxides, nickel, zinc, manganese, or oxides, alloys, or other combinations of these or other materials having specified magnetic properties.

(43) The magnetic shield **270** additionally includes two mounting holes **273a**, **273b** configured to facilitate the assembly of the magnetic shield **270** and other elements of the wearable device **200**. For example, screws, bolts, or other fasteners could pass through the mounting holes **273a**, **273b** to attach elements of the wearable device **200** (e.g., the housing **210**, the circuit board **280**, the interface **290**) to other elements of the wearable device **200** (e.g., electrical contacts **229a**, **249b** used to detect a GSR of skin of a wearer and/or to enable other applications of the wearable device **200**). Other configurations of the magnetic shield **270** and/or additional or alternative magnetic elements (e.g., magnetic cores configured to focus flux through the windings of the coil **250**, ferrite cans, laminated sheets of magnetic material configured to minimize eddy current losses) are anticipated. For example, the magnetic shield **270** and/or other magnetic materials of the wearable device **200** could be magnetized or have some other magnetic property such that the wearable device **200** experiences an aligning magnetic force when mounted on a wireless charging device.

(44) The flexible PCB **260** and components disposed thereupon could be configured to increase the efficiency of electromagnetic energy reception by the coil **250**. For example, very few components could be disposed on the flexible PCB **260**. For example, the wearable device may include a blood oxygenation and pulse oximetry sensor that includes a photodiode **240** and two light-emitting diodes (LEDs) **242, 244**. Amplifiers, current sources, controllers, ADCs, and other components of the blood oxygenation and pulse oximetry sensor could be disposed on the circuit board **280** on the opposite side of the magnetic shield **270** from the coil **250** while only elements of the sensor requiring direct access to the body of a wearer (e.g., the photodiode **240** and LEDs **242, 244** and a minimum of metallic tracing patterned on the flexible PCB **260** to electrically couple the photodiode **240** and LEDs **242, 244** to the other components of the sensor) are disposed on the flexible PCB **260**. In general, a minimum of conductive material could be disposed on the flexible PCB **260** (e.g., printed circuit traces, electronic components). In some examples, regions of the flexible PCB **270** that do not feature traces could not include conductive material. For example, the flexible PCB **270** could lack ground planes, shield planes, signal or other large ‘pours’ or other large contiguous regions wholly or partially covered with conductive material.

(45) The windings of the coil **250**, when assembled into the wearable device **200**, are disposed proximate to a peripheral portion of the contact surface **230** of the housing **210**. The coil **250** could be configured in this way to maximize the area enclosed by the coil **250** while remaining disposed within the housing **210**. The windings of the coil **250** could be configured differently according to an application. For example, the coil **250** could include figure-eight windings, multiple discrete sets of windings, windings having a rectangular shape, windings having an elliptical shape, or other patterns, shapes, or configurations according to an application. In some examples the coil **250** could be disposed on the flexible PCB **260**. For example, one or more windings of the coil **250** could be a trace on the flexible PCB **260**. One or more properties of the coil **250** could be specified according to an application. In some examples, the coil **250** could have a substantially rectangular shape and a size of approximately 30 millimeters by 22 millimeters, corresponding to an internal shape and size of the periphery of the housing **210** of the wearable device **200**. The shape, size, number of windings, and other properties of the coil, **250** (as well as properties of components of a recharger powered by electromagnetic energy received by the coil **250**) could be specified such that the coil is able to receive electromagnetic energy having a specified frequency. The specified frequency could be in the range of 100 kilohertz to 200 kilohertz. The specified frequency could be specified by a wireless charging standard or standards (e.g., the Qi wireless charging standard).

(46) Sensors of the wearable device configured to detect one or more properties the body of a wearer of the wearable device **200** could include a variety of components and could function using a variety of different mechanisms. The sensors could include light sensors, sound sensors, vibration sensors, electrical sensors (e.g., current sensors, electric field sensors, voltage sensors), electrical contacts or probes, magnetic sensors, electromagnetic energy sensors, acoustic sensors, accelerometers, pressure sensors, IR sensors, cameras, temperature sensors, or other sensors or combinations of sensors. Further, the sensors could be active sensors or could otherwise include energy emitters, including but not limited to light emitters, LEDs, lasers, electromagnetic energy emitters, emitter antennas, emitter coils, microwave emitters, magnetic field emitters, magnets, IR emitters, UV emitters, vibrators, or other energy emitting elements or combinations of energy emitting elements.

(47) The sensors could be operated to detect one or more of a variety of properties of a wearer of the wearable device **200** via an external body surface of the wearer e.g., by contacting, directing energy (e.g., electrical, magnetic, illumination, acoustic waves) through/into, detecting energy (e.g., electrical, magnetic, illumination, acoustic waves) received from/through, or otherwise interacting with and/or through an external body surface of the wearer (i.e., skin of the wearer's wrist). For example, light could be emitted (e.g., using the LEDs **242, 244**) toward an external body surface of a wearer to illuminate the external body surface, and one or more properties of light received from

the external body surface could be detected (e.g., using the photodiode). This illumination and detection could be used to detect an oxygenation state of blood proximate to the wearable device (e.g., in the skin of the external body surface), a heart rate of the wearer, a flow profile of the blood in vasculature of the wearer, or some other information. The sensors could be configured to detect one or more properties of a contrast agent (e.g., a functionalized fluorophore, chromophore, magnetic particle, or some other natural or artificial contrast agent) in the body of the wearer proximate to the wearable device **200** according to an application.

(48) In some examples, contacts could protrude from the wearable device **200** and could facilitate physical measurement of properties of the body of the wearer (e.g., of the skin at the external body surface). FIG. 2D illustrates a view of the contact surface **230** of the wearable device **200**, including two circular, rounded contacts **249a**, **249b** configured to protrude from the contact surface **230** and to make electrical and/or thermal contact with skin of the external body surface of the wearer. The contacts **249a**, **249b** are in electrical contact with electrical pads **248a**, **248b**, respectively, on the flexible PCB **260** such that a GSR of the skin at the external body surface can be detected by the wearable device **200** (e.g., by using components of the electronics **282** to apply a specified voltage between the contacts **249a**, **249b** and detecting a current through the contacts **249a**, **249b** related to the GSR of the skin proximate to the contacts **249a**, **249b**). One of the contacts **249a** is additionally in thermal contact with a thermometer **246** disposed on the flexible PCB **260** such that the temperature at the external body surface of a wearer could be detected by the thermometer **246** using the first contact **249a**. Other uses and configurations of electrical, thermal, or other physical contacts (e.g., **249a**, **249b**) are anticipated.

(49) One or more components of the sensors could be disposed on the contact surface **230** of the housing **210** such that the one or more components could access (e.g., detect a property of, emit energy toward, illuminate, physically contact) an external body surface of a wearer. The one or more components could be disposed proximate to a window, filter, grate, hole, or other feature of the housing **210** to facilitate operate of the sensor. The housing **210** includes a window **235** through which the LEDs **242**, **244** can illuminate skin of the external body surface of the wearer. The photodiode **240** is also positioned proximate to the window **235** to enable the photodiode **240** to receive light from the external body surface. The window **235** could be fitted with a wholly or partially transparent window or other filter to enable the housing **210** to be water-resistant and/or water-proof. Additionally or alternatively, the housing **210** could be configured to include sealants, adhesives, gaskets, welds, press-fitted seams, and/or other joints such that the housing **210** was resistant to water entering an internal volume or volumes of the housing **210** when the housing **210** is exposed to water. The housing **210** could further be water-proof, i.e., resistant to water entering an internal volume or volumes of the housing **210** when the housing **210** is submerged in water. For example, the housing **210** could be water-proof to a depth of 1 meter, i.e., configured to resist water entering an internal volume or volumes of the housing **210** when the housing **210** is submerged to a depth of 1 meter. Further, the interface between the housing **210** and other elements of the wearable device **200** (e.g., elements of a sensor, buttons, user interface elements, electrical contacts) protruding from, embedded in the surface of, or otherwise interrupting the material of the housing **210** could be configured such that the combination of the housing **210** and the other elements of the wearable device **200** is water-resistant and/or water-proof.

(50) Note that the embodiments illustrated in FIGS. 2A-D are illustrative examples and not meant to be limiting. Alternative embodiments, including more or fewer components in alternative configurations are anticipated. A wearable device could include multiple housings or other such assemblies each containing some set of components to enable applications of such a wearable device. For example, a wearable device could include a first housing within which are disposed a coil configured to received electromagnetic energy and a sensor configured to detect one or more properties of a wearer and a second housing containing a rechargeable battery and electronics configured to recharge the rechargeable battery using energy received using the coil. Wearable

device could be configured to perform a variety of functions and to enable a variety of applications. Wearable devices could be configured to operate in concert with other devices or systems; for example, wearable devices could include a wireless communication interface configured to transmit data indicative of one or more properties of the body of a wearer of the wearable device. Other embodiments, operations, configurations, and applications of a wearable device as described herein are anticipated.

III. Example Devices

(51) A wearable device **300** (illustrated in FIG. 3) can automatically measure a plurality of physiological parameters of a person wearing the device. The term “wearable device,” as used in this disclosure, refers to any device that is capable of being worn at, on or in proximity to a body surface, such as a wrist, ankle, waist, chest, or other body part. In order to take in vivo measurements in a non-invasive manner from outside of the body, the wearable device may be positioned on a portion of the body where subsurface vasculature or other elements of the body of the wearer are easily observable, the qualification of which will depend on the type of detection system used. The device may be placed in close proximity to the skin or tissue. A mount **310**, such as a belt, wristband, ankle band, etc. can be provided to mount the device at, on or in proximity to the body surface. The mount **310** may prevent the wearable device from moving relative to the body to reduce measurement error and noise. In one example, shown in FIG. 3, the mount **310**, may take the form of a strap or band **320** that can be worn around a part of the body. Further, the mount **310** may be an adhesive substrate for adhering the wearable device **300** to the body of a wearer.

(52) A housing **330** is disposed on the mount **310** such that it can be positioned on the body. A contact surface **340** of the housing **330** is intended to be mounted facing to the external body surface. The housing **330** may include at least one sensor **380** for detecting at least one property of the body of the wearer, which could include any parameters that may relate to the health of the person wearing the wearable device. For example, the sensor **380** could be configured to measure blood pressure, pulse rate, respiration rate, skin temperature, etc. In a non-exhaustive list, sensor **380** may include any one of an optical (e.g., CMOS, CCD, photodiode), acoustic (e.g., piezoelectric, piezoceramic), electrochemical (voltage, impedance), thermal, mechanical (e.g., pressure, strain), magnetic, or electromagnetic (e.g., magnetic resonance) sensor. Components disposed in the housing **330** may be miniaturized so that the wearable device may be worn on the body without significantly interfering with the wearer's usual activities.

(53) The housing **330** additionally encloses a coil (not shown) configured to received electromagnetic energy. The coil could be disposed proximate to the contact surface **340** of the housing **330** such that the coil could efficiently receive electromagnetic energy emitted by a wireless charger (not shown) when the wearable device **300** is mounted on such a wireless charger. The wearable device additionally includes a rechargeable battery (not shown) configured to power the wearable device **300**. A recharger (not shown) of the wearable device **300** is configured to recharge the rechargeable battery using electromagnetic energy received using the coil. The wearable device additionally includes electronics configured to enable functions of the wearable device **300** including operating the sensor **380** to detect one or more properties of the body of a wearer.

(54) The wearable device **300** may also include a user interface **390** via which the wearer of the device may receive one or more recommendations or alerts generated either from a remote server or other remote computing device, or from a processor within the device. The alerts could be any indication that can be noticed by the person wearing the wearable device. For example, the alert could include a visual component (e.g., textual or graphical information on a display), an auditory component (e.g., an alarm sound), and/or tactile component (e.g., a vibration). Further, the user interface **390** may include a display **392** where a visual indication of the alert or recommendation may be displayed. The display **392** may further be configured to provide an indication of the

measured physiological parameters, for instance, the concentrations of certain blood analytes being measured.

(55) In some examples, the wearable device is provided as a wrist-mounted device, as shown in FIGS. 4A and 4B. The wrist-mounted device may be mounted to the wrist of a living subject with a wristband or cuff, similar to a watch or bracelet. As shown in FIGS. 4A and 4B, the wrist mounted device **400** may include a mount **410** in the form of a wristband **420**, a housing **430** positioned on the anterior side **440** of the wearer's wrist, and a user interface **450** positioned on the posterior side **460** of the wearer's wrist. The wearer of the device may receive, via the user interface **450**, one or more recommendations or alerts generated either from a remote server or other remote computing device, or alerts from the measurement platform. Such a configuration may be perceived as natural for the wearer of the device in that it is common for the posterior side **460** of the wrist to be observed, such as the act of checking a wrist-watch. Accordingly, the wearer may easily view a display **470** on the user interface. Further, the housing **430** may be located on the anterior side **440** of the wearer's wrist where the subsurface vasculature or other elements of the body of the wearer may be readily observable. However, other configurations are contemplated.

(56) The display **470** may be configured to display a visual indication of the alert or recommendation and/or an indication of the measured physiological parameters, for instance, the presence or concentrations of certain blood analytes being measured. Further, the user interface **450** may include one or more buttons **480** for accepting inputs from the wearer. For example, the buttons **480** may be configured to change the text or other information visible on the display **470**. As shown in FIG. 4B, housing **430** may also include one or more buttons **490** for accepting inputs from the wearer. The buttons **490** may be configured to accept inputs for controlling aspects of the data collection system, such as initiating a measurement period, or inputs indicating the wearer's current health state (i.e., normal, migraine, shortness of breath, heart attack, fever, "flu-like" symptoms, food poisoning, etc.).

(57) FIG. 5 is a simplified schematic of a system including one or more wearable devices **500**. The one or more wearable devices **500** may be configured to transmit data via a communication interface **510** over one or more communication networks **520** to a remote server **530**. In one embodiment, the communication interface **510** includes a wireless transceiver for sending and receiving communications to and from the server **530**. In further embodiments, the communication interface **510** may include any means for the transfer of data, including both wired and wireless communications. For example, the communication interface may include a universal serial bus (USB) interface or a secure digital (SD) card interface. Communication networks **520** may be any one of may be one of: a plain old telephone service (POTS) network, a cellular network, a fiber network and a data network. The server **530** may include any type of remote computing device or remote cloud computing network. Further, communication network **520** may include one or more intermediaries, including, for example wherein the wearable device **500** transmits data to a mobile phone or other personal computing device, which in turn transmits the data to the server **530**.

(58) In addition to receiving communications from the wearable device **500**, such as collected physiological parameter data and data regarding health state as input by the user and/or one or more properties of a wearer detected using a sensor disposed in the wearable device **500**, the server may also be configured to gather and/or receive either from the wearable device **500** or from some other source, information regarding a wearer's overall medical history, environmental factors and geographical data. For example, a user account may be established on the server for every wearer that contains the wearer's medical history. Moreover, in some examples, the server **530** may be configured to regularly receive information from sources of environmental data, such as viral illness or food poisoning outbreak data from the Centers for Disease Control (CDC) and weather, pollution and allergen data from the National Weather Service. Further, the server may be configured to receive data regarding a wearer's health state from a hospital or physician. Such information may be used in the server's decision-making process, such as recognizing correlations

and in generating clinical protocols.

(59) Additionally, the server may be configured to gather and/or receive the date, time of day and geographical location of each wearer of the device during each measurement period. Such information may be used to detect and monitor spatial and temporal spreading of diseases. As such, the wearable device may be configured to determine and/or provide an indication of its own location. For example, a wearable device may include a GPS system so that it can include GPS location information (e.g., GPS coordinates) in a communication to the server. As another example, a wearable device may use a technique that involves triangulation (e.g., between base stations in a cellular network) to determine its location. Other location-determination techniques are also possible.

(60) The server may also be configured to make determinations regarding the efficacy of a drug or other treatment based on information regarding the drugs or other treatments received by a wearer of the device and, at least in part, the physiological parameter data and the indicated health state of the user. From this information, the server may be configured to derive an indication of the effectiveness of the drug or treatment. For example, if a drug is intended to treat nausea and the wearer of the device does not indicate that he or she is experiencing nausea after beginning a course of treatment with the drug, the server may be configured to derive an indication that the drug is effective for that wearer. In another example, a wearable device may be configured to measure blood glucose. If a wearer is prescribed a drug intended to treat diabetes, but the server receives data from the wearable device indicating that the wearer's blood glucose has been increasing over a certain number of measurement periods, the server may be configured to derive an indication that the drug is not effective for its intended purpose for this wearer.

(61) Further, some embodiments of the system may include privacy controls which may be automatically implemented or controlled by the wearer of the device. For example, where a wearer's collected physiological parameter data and health state data are uploaded to a cloud computing network for trend analysis by a clinician, the data may be treated in one or more ways before it is stored or used, so that personally identifiable information is removed. For example, a user's identity may be treated so that no personally identifiable information can be determined for the user, or a user's geographic location may be generalized where location information is obtained (such as to a city, ZIP code, or state level), so that a particular location of a user cannot be determined.

(62) Additionally or alternatively, wearers of a device may be provided with an opportunity to control whether or how the device collects information about the wearer (e.g., information about a user's medical history, social actions or activities, profession, a user's preferences, or a user's current location), or to control how such information may be used. Thus, the wearer may have control over how information is collected about him or her and used by a clinician or physician or other user of the data. For example, a wearer may elect that data, such as health state and physiological parameters, collected from his or her device may only be used for generating an individual baseline and recommendations in response to collection and comparison of his or her own data and may not be used in generating a population baseline or for use in population correlation studies.

IV. Example Electronics Platform for a Device

(63) FIG. 6 is a simplified block diagram illustrating the components of a device **600**, according to an example embodiment. Device **600** may take the form of or be similar to one of the wearable devices **100**, **200**, **300**, **400**, shown in FIGS. 1A-C, 2A-D, 3, and 4A-B. However, device **600** may also take other forms, such as an ankle, waist, or chest-mounted device. Device **600** could also take the form of a device that is not configured to be mounted to a body. For example, device **600** could take the form of a handheld device configured to be maintained in proximity to an environment of interest (e.g., a body part, a biological sample container, a volume of a water treatment system) by a user or operator of the device **600** or by a frame or other supporting structure. Device **600** also

could take other forms.

(64) In particular, FIG. 6 shows an example of a device **600** having a sensor **612**, a coil **614**, a power supply **616**, a rechargeable battery **618**, a user interface **620**, communication interface **630** for transmitting data to a remote system, and processor(s) **640**. The components of the device **600** may be disposed on a mount or on some other structure for mounting the device to enable stable detection of one or more properties of an environment of interest (e.g., of a body of a wearer of the device **600**), for example, to an external body surface where a portion of subsurface vasculature or other anatomical element is readily observable.

(65) Processor **640** may be a general-purpose processor or a special purpose processor (e.g., digital signal processors, application specific integrated circuits, etc.). The one or more processors **640** can be configured to execute computer-readable program instructions **670** that are stored in the computer readable medium **660** and that are executable to provide the functionality of a device **600** described herein.

(66) The computer readable medium **660** may include or take the form of one or more non-transitory, computer-readable storage media that can be read or accessed by at least one processor **640**. The one or more computer-readable storage media can include volatile and/or non-volatile storage components, such as optical, magnetic, organic or other memory or disc storage, which can be integrated in whole or in part with at least one of the one or more processors **640**. In some embodiments, the computer readable medium **660** can be implemented using a single physical device (e.g., one optical, magnetic, organic or other memory or disc storage unit), while in other embodiments, the computer readable medium **660** can be implemented using two or more physical devices.

(67) Sensor **612** could include a component configured to detect one or more properties of an environment proximate to the sensor **612** (e.g., skin of an external body surface of a wearer of the device **600**) and/or of energy or matter received from the proximate environment. As described above, the sensor **612** may include any component or components capable of detecting at least one property, which could include any properties that may relate to the environment being analyzed by the device (e.g., the body of the wearer or a subsection thereof). For example, the sensor **612** could be configured to measure blood pressure, pulse rate, skin temperature, etc. In some examples, the sensor **612** may include one or more of an optical (e.g., CMOS, CCD, photodiode), acoustic (e.g., piezoelectric, piezoceramic), electrochemical (voltage, impedance), thermal, mechanical (e.g., pressure, strain), magnetic, or electromagnetic (e.g., magnetic resonance) sensor. In examples wherein the sensor **612** includes a light sensor, the light sensor could be a photodiode, a photomultiplier, a CCD, a photocell, a photoresistive element, a camera, or any other sensor or sensors configured to detect one or more properties of light emitted by color centers of the functionalized nanodiamonds.

(68) The sensor **612** could additionally include a light source or other energy emitter for transmitting illumination or other energy that can illuminate and/or penetrate the environment to illuminate, excite, or otherwise affect one or more elements of interest in the environment proximate to the sensor **612** (e.g., a fluorescent contrast agent configured to bind to an analyte of interest in blood of a wearer of the device **600**). The wavelength of transmitted illumination could be specified to penetrate biological tissues of a wearer; for example, the transmitted illumination could have a wavelength within a near-infrared (NIR) transparency window of biological tissue. The wavelength of the transmitted illumination could be specified to be a wavelength that causes fluorescence and/or emission of light by fluorophores, chromophores, or other elements of interest. An energy emitter of the sensor **612** could be configured to produce other forms of energy toward the environment proximate to the sensor **612** that could result in emission, reflection, scattering or some other generation of light or other energy or matter by other chemicals, imaging agents, biological elements, or other analytes proximate to the sensor **612**.

(69) Coil **614** comprises windings that are configured to receive electromagnetic energy. The coil

614 is shaped such that the windings of the coil **614** enclose a region of the device **600** wherein the sensor **612** is disposed. The coil **614** can be configured in a number of ways to enable efficient reception of electromagnetic energy using the coil **614** or to enable and/or facilitate a number of other applications. The windings of the coil **614** could be disposed proximate to a peripheral portion of a contact surface of a housing of the device **600** such that an area enclosed by the coil **614** (e.g., a central portion of the contact surface of the housing) is maximized and/or such that a separation distance between the coil **614** and a charging coil of a wireless charger is minimized. The coil **614** could have a rectangular shape, an elliptical shape, or some other shape according to an application; for example, the shape of the coil **614** could correspond to the shape of the contact surface of the housing of the device **600**. The coil **614**, and/or other components could be configured to enable efficient reception of electromagnetic energy of a specific frequency (e.g., 100 kilohertz to 200 kilohertz) by the coil **614**. For example, the coil and a capacitor coupled to the coil **614** could be configured to have a resonant frequency equal to the specific frequency of the electromagnetic energy.

(70) Rechargeable battery **618** is configured to power the device **600** using stored electrochemical energy and to be recharged a plurality of times. The rechargeable battery **618** could include one or more of a variety of rechargeable battery chemistries, including lead-acid, nickel-metal-hydride, nickel-cadmium, lithium-ion, lithium-polymer, or some other rechargeable battery chemistry. Power supply **616** is configured to recharge the rechargeable battery **618** by applying a constant current, a constant voltage, a trickle current, or some other electrical energy having one or more specified properties to two or more electrodes of the rechargeable battery **618**. The rechargeable battery **618** could include one or more thermistors that the processor(s) **640**, the power supply **616**, or some other component of the device **600** could operate to determine a temperature of the rechargeable battery **618** and to prevent damage of the rechargeable battery **618** by reducing a charging rate, a discharging rate, or some other property of use of the rechargeable battery **618** to prevent damage of the rechargeable battery **618**.

(71) The power supply **616** is configured to provide power to the device **600** from the rechargeable battery **618** and to recharge the rechargeable battery **618** using electromagnetic energy received using the coil **614**. The power supply **616** could include components configured to facilitate the reception of electromagnetic energy by the coil **614**. For example, the power supply **616** could include a fixed and/or variable capacitor configured to act as part of an LC tank circuit with the coil **614** such that the LC tank circuit had a resonant frequency substantially equal to a frequency of electromagnetic energy directed toward the coil. In some examples, the power supply **616** could be configured to load the coil **614** or otherwise alter one or more properties of the coil and/or power supply **616** in order to use the coil **614** to communicate with an external system. For example, the power supply **616** could alter a load presented to the coil **614** such that the coil **614** could reflect backscatter electromagnetic radiation toward a wireless charger in order to communicate a charge state, a required power level, or some other property of the coil **614** and/or device **600**.

(72) The program instructions **670** stored on the computer readable medium **660** may include instructions to perform any of the methods described herein. For instance, in the illustrated embodiment, program instructions **670** include a controller module **672**, calculation and decision module **674** and an alert module **676**.

(73) The controller module **672** can include instructions for operating the sensor **612**. For example, the controller **672** may operate a light source and/or light sensors of the sensor **612** during each of a set of pre-set measurement periods. The controller module **672** can include instructions for operating the power supply **616** to recharge the rechargeable battery **618** using electromagnetic energy received using the coil **614**, to operate the coil **614** to transmit information to another system (e.g., a wireless charger), to route power between the coil **614**, the rechargeable battery **618**, and other components of the device **600**, or to perform other functions. For example, the controller module **672** can include instructions for operating the power supply **616** to apply a pattern of loads

to the coil **614** in order to reflect backscatter electromagnetic radiation reflect backscatter electromagnetic radiation toward a wireless charger in order to communicate a charge state, a required power level, or some other property of the coil **614** and/or device **600** (e.g., to communicate information and/or operate the rechargeable battery **618** according to a wireless power transfer standard, e.g., the Qi wireless power transfer standard).

(74) The controller module **672** can also include instructions for operating a user interface **620**. For example, controller module **672** may include instructions for displaying data collected by the data collection system **610** and analyzed by the calculation and decision module **674**, or for displaying one or more alerts generated by the alert module **676**. Further, controller module **672** may include instructions to execute certain functions based on inputs accepted by the user interface **620**, such as inputs accepted by one or more buttons disposed on the user interface.

(75) Communication platform **630** may also be operated by instructions within the controller module **672**, such as instructions for sending and/or receiving information via a wireless antenna, which may be disposed on or in the device **600**. The communication interface **630** can optionally include one or more oscillators, mixers, frequency injectors, etc. to modulate and/or demodulate information on a carrier frequency to be transmitted and/or received by the antenna. In some examples, the device **600** is configured to indicate an output from the processor by modulating an impedance of the antenna in a manner that is perceivable by a remote server or other remote computing device.

(76) Calculation and decision module **672** may include instructions for receiving data from the sensor **612**, analyzing the data to determine one or more properties of an environment proximate to the sensor **612** (e.g., of a body of a wearer of the device **600**), such as concentration of a target analyte, analyzing the data to determine if a medical condition or other specified condition is indicated, or other analytical processes relating to the environment proximate to the device **600**. In particular, the calculation and decision module **672** may include instructions for determining, for each preset measurement time, the presence, concentration, and/or other properties of a clinically-relevant analyte based on the one or more properties of light emitted by contrast agents in a lumen of subsurface vasculature of a user of the device **600**; and determining whether a medical condition is indicated based on at least the corresponding presence, concentration, or other property of the clinically-relevant analyte. These instructions could be executed at each of a set of preset measurement times.

(77) The program instructions of the calculation and decision module **672** may, in some examples, be stored in a computer-readable medium and executed by a processor located external to the device **600**. For example, the device **600** could be configured to collect certain data regarding physiological parameters from the user and then transmit the data to a remote server, which may include a mobile device, a personal computer, the cloud, or any other remote system, for further processing.

(78) The computer readable medium **660** may further contain other data or information, such as medical and health history of a user of the device **600**, that may be useful in determining whether a medical condition or some other specified condition is indicated. Further, the computer readable medium **660** may contain data corresponding to certain physiological parameter baselines, above or below which a medical condition is indicated. The baselines may be pre-stored on the computer readable medium **660**, may be transmitted from a remote source, such as a remote server, or may be generated by the calculation and decision module **674** itself. The calculation and decision module **674** may include instructions for generating individual baselines for the user of the device **600** based on data collected over a certain number of measurement periods. For example, the calculation and decision module **674** may generate a baseline concentration of a pulse rate and galvanic skin resistance for each of a plurality of measurement periods by averaging the pulse rate and galvanic skin resistance at each of the measurement periods measured over the course of a few days, and store those baseline pulse rates and galvanic skin resistances in the computer readable

medium **660** for later comparison. Baselines may also be generated by a remote server and transmitted to the device **600** via communication interface **630**. The calculation and decision module **674** may also, upon determining that a medical or other emergency condition is indicated, generate one or more recommendations for the user of the device **600** based, at least in part, on consultation of a clinical protocol. Such recommendations may alternatively be generated by the remote server and transmitted to the device **600**.

(79) In some examples, the collected physiological parameter data, baseline profiles, health state information input by device users and generated recommendations and clinical protocols may additionally be input to a cloud network and be made available for download by a user's physician. Trend and other analyses may also be performed on the collected data, such as physiological parameter data and health state information, in the cloud computing network and be made available for download by physicians or clinicians.

(80) Further, physiological parameter and health state data from individuals or populations of device users may be used by physicians or clinicians in monitoring efficacy of a drug or other treatment. For example, high-density, real-time data may be collected from a population of device users who are participating in a clinical study to assess the safety and efficacy of a developmental drug or therapy. Such data may also be used on an individual level to assess a particular wearer's response to a drug or therapy. Based on this data, a physician or clinician may be able to tailor a drug treatment to suit an individual's needs.

(81) In response to a determination by the calculation and decision module **674** that a medical or other specified condition is indicated, the alert module **676** may generate an alert via the user interface **620**. The alert may include a visual component, such as textual or graphical information displayed on a display, an auditory component (e.g., an alarm sound), and/or tactile component (e.g., a vibration). The textual information may include one or more recommendations, such as a recommendation that the user of the device contact a medical professional, seek immediate medical attention, or administer a medication.

CONCLUSION

(82) Where example embodiments involve information related to a person or a device of a person, the embodiments should be understood to include privacy controls. Such privacy controls include, at least, anonymization of device identifiers, transparency and user controls, including functionality that would enable users to modify or delete information relating to the user's use of a product.

(83) Further, in situations in where embodiments discussed herein collect personal information about users, or may make use of personal information, the users may be provided with an opportunity to control whether programs or features collect user information (e.g., information about a user's medical history, social network, social actions or activities, profession, a user's preferences, or a user's current location), or to control whether and/or how to receive content from the content server that may be more relevant to the user. In addition, certain data may be treated in one or more ways before it is stored or used, so that personally identifiable information is removed. For example, a user's identity may be treated so that no personally identifiable information can be determined for the user, or a user's geographic location may be generalized where location information is obtained (such as to a city, ZIP code, or state level), so that a particular location of a user cannot be determined. Thus, the user may have control over how information is collected about the user and used by a content server.

(84) The particular arrangements shown in the Figures should not be viewed as limiting. It should be understood that other embodiments may include more or less of each element shown in a given Figure. Further, some of the illustrated elements may be combined or omitted. Yet further, an exemplary embodiment may include elements that are not illustrated in the Figures.

(85) Additionally, while various aspects and embodiments have been disclosed herein, other aspects and embodiments will be apparent to those skilled in the art. The various aspects and embodiments disclosed herein are for purposes of illustration and are not intended to be limiting, with the true

scope and spirit being indicated by the following claims. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented herein. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are contemplated herein.

Claims

1. A wearable device, comprising: a housing, wherein the housing includes a contact surface for contacting an external body surface; a physiological sensor mounted on a first side of a flexible circuit board disposed within the housing; electronics disposed within the housing, wherein the electronics comprise a controller that is configured to operate the physiological sensor to detect one or more physiological parameters via the external body surface; a coil configured to receive electromagnetic energy for powering the electronics, wherein the coil is disposed within the housing proximate to the contact surface of the housing, wherein the coil comprises a plurality of windings that surround the physiological sensor; a magnetic shield disposed within the housing between the coil and at least one electronic component of the electronics, wherein the magnetic shield shields the at least one electronic component from electromagnetic energy, wherein a second side of the flexible circuit board, which is opposite the first side of the flexible circuit board upon which the physiological sensor is disposed, along with the coil are both mounted on the magnetic shield, wherein the flexible circuit board passes through a slot in the magnetic shield to provide an electrical connection between the electronics and the physiological sensor, and wherein the physiological sensor and the coil are both mounted on a common side of the magnetic shield while the controller that is configured to operate the physiological sensor is disposed on an opposite side of the magnetic shield; and a battery disposed within the housing and positioned between the electronics and the magnetic shield.
2. The wearable device of claim 1, wherein the battery comprises a rechargeable battery.
3. The wearable device of claim 2, wherein the electronics further comprises: a recharger, wherein the recharger is powered by electromagnetic energy received by the coil to recharge the rechargeable battery.
4. The wearable device of claim 1, wherein the magnetic shield is configured to focus electromagnetic energy directed toward the coil.
5. The wearable device of claim 1, wherein the magnetic shield is configured to increase a coupling between the coil and an external charging coil that is directing electromagnetic energy toward the coil.
6. The wearable device of claim 1, wherein the controller is further configured to operate the coil to transmit information to a wireless charger.
7. The wearable device of claim 6, wherein the information comprises an indication of an amount of electromagnetic energy to emit toward the coil.
8. The wearable device of claim 7, wherein the indication is provided by changing an electromagnetically detectable property of the coil in a pattern.
9. The wearable device of **8**, wherein the electromagnetically detectable property of the coil comprises an impedance of the coil.
10. The wearable device of claim 1, wherein the physiological sensor is disposed proximate to a window or hole in the housing.
11. The wearable device of claim 1, further comprising: a magnetic material that exerts an aligning magnetic force that aligns the wearable device with a wireless charger.
12. The wearable device, of claim 11, wherein the magnetic material comprises one or more permanent magnets.

13. The wearable device of claim 1, wherein the physiological sensor comprises one or light emitters configured to emit light toward the external body surface and a light sensor configured to detect one or more properties of light received from the external body surface.
 14. The wearable device of claim 13, wherein the controller is configured to determine a heart rate from the one or more properties of light detected by the light sensor.
 15. The wearable device of claim 1, further comprising at least two contacts to measure properties of the external body surface.
 16. The wearable device of claim 1, wherein the coil has a substantially rectangular shape.
 17. The wearable device of claim 1, further comprising a user interface configured to provide a user-discernible indication of the one or more physiological parameters.
 18. The wearable device of claim 1, further comprising a wireless communication interface configured to transmit data indicative of the one or more physiological parameters.
 19. The wearable device of claim 1, wherein the physiological sensor is disposed proximate to a window or a hole in the housing and wherein the windings of the coil surround the flexible circuit board and the window or the hole in the housing.
 20. The wearable device of claim 19, wherein the windings of the coil surround the slot in the magnetic shield where the flexible interconnected passes through the magnetic shield.
 21. The wearable device of claim 1, wherein the magnetic shield extends beyond perimeters of the flexible circuit board and the windings of the coil.
 22. The wearable device of claim 1, wherein the electronics are disposed within the housing on a second circuit board disposed on an opposite side of the magnetic shield as the flexible circuit board, wherein the magnetic shield shields the second circuit board from the electromagnetic energy.
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